European Commission - Press release

Mergers: Commission clears acquisition of VMware by Broadcom, subject to conditions

Brussels, 12 July 2023

The European Commission has approved, under the EU Merger Regulation, the proposed acquisition of VMware by Broadcom. The approval is conditional upon full compliance with the commitments offered by Broadcom.

Today's decision follows an in-depth investigation of the proposed acquisition. **Broadcom** is a hardware company that offers, among other products, Fibre Channel Host-Bus Adapters (‘FC HBAs'), storage adapters and Network Interface Cards (‘NICs'), which are hardware components that connect servers to storage or network. Broadcom has recently started expanding into software markets, mainly for security and mainframe applications. **VMware** is a software supplier offering mainly virtualization software that interoperates with a wide range of hardware, including FC HBAs, storage adapters and NICs.

The Commission's investigation

The Commission's preliminary market investigation found that, by acquiring VMware, Broadcom could have restricted competition in the markets for the supply of NICs, FC HBAs and storage adapters.

The Commission's in-depth investigation confirmed that the transaction, as initially notified, **would harm competition in the worldwide market for the supply of FC HBAs**. At the same time, it ruled out possible competition concerns related to: (i) the market for the supply of NICs and storage adapters, (ii) the market for the development of SmartNICs; and (iii) the bundling of Broadcom's virtualisation software with VMware's.

In particular, the Commission found that:

- Broadcom would have the **ability and incentive to foreclose Marvell**, the only rival on the market for the supply of FC HBAs, by restricting or degrading the interoperability between VMware's server virtualisation software and Marvell's hardware. This would hamper Marvell's ability to compete in a market where Broadcom is dominant, or at least holds a very strong position, ultimately leading to higher prices, lower quality and less innovation for business customers.
- Broadcom **does not hold a strong position** in the market for storage adapters and NICs and will **continue to face competition** from several rival suppliers.
- Broadcom would have **no economic incentive to hinder** the development of **SmartNICs** by other **providers** by decreasing VMware's involvement in Project Monterey, an ongoing cooperation with three other SmartNICs sellers (NVIDIA, Intel and AMD Pensando), as such foreclosure would not be profitable.
- Broadcom would **not be able to bundle VMware's virtualisation software with its own software** (namely mainframe and security software) as such products are purchased by different divisions in a customer organization and/or at different points in time.

The remedies

To address the Commission's competition concerns in the worldwide market for the supply of FC HBAs, Broadcom offered the following comprehensive **access and interoperability commitments** to Marvell and to any potential future entrant:

- Guaranteed **access to the interoperability Application Programming Interfaces** as well as to the **materials, tools and technical support** necessary for the development and certification of third-party FC HBAs. It also committed to ensuring (i) their interoperability with VMware's server virtualisation software; and (ii), third parties' access to this information at the same time as Broadcom.
- Guaranteed **access to the source code** for all of Broadcom's current and future FC HBA

drivers through an irrevocable open source license. This would allow Marvell and any potential entrant to ensure interoperability with VMware's server virtualisation software and allow them to reuse and modify Broadcom's drivers for its own use.

Furthermore, Broadcom committed to implementing an **organisational separation** between the team working on Broadcom's FC HBAs and the team in charge of third-party certification and technical support. It also committed to ensuring protection of confidential information of Marvell and any potential entrant obtained in the context of the interoperability and certification processes.

The Commission carefully investigated the effectiveness of the remedies, collecting views from Marvell as well as server manufacturers, which are the main direct customers of FC HBAs. In view of the **positive feedback from market participants**, the Commission concluded that the **proposed acquisition, as modified by the commitments, would no longer raise competition concerns** and would maintain competition on the market for FC HBAs.

The Commission's decision is conditional upon full compliance with the commitments, which will be in place for ten years. Under supervision of the Commission, an independent trustee will be in charge of monitoring compliance with the commitments. Moreover, a **fast-track dispute resolution mechanism** in case of disagreement between the parties will further ensure the commitments' effective implementation.

Companies and products

Broadcom is a US based hardware company active in the production of NICs, FC HBAs and storage adapters, amongst other products. NICs are server components that provide an interface between the server and other computers and equipment on a network. FC HBAs are storage adapters that connect servers to storage located outside the server on a storage-area network using the fibre channel protocol, typically through a switch. Storage adapters connect servers' central processing units to storage directly. Broadcom recently started expanding into software markets, notably through the acquisition of CA Technology and Symantec.

VMware is a US based software company that mainly offers virtualisation software, which enables IT operators to create virtual versions of computer resources (e.g. hardware, operating systems, storage devices, or network resources) known as virtual machines. This enables IT organizations to run more than one virtual system – and multiple OS and applications – on a single server at the same time.

Merger control rules and procedure

The Commission has the duty to assess mergers and acquisitions involving companies with a turnover above certain thresholds (see Article 1 of the EU Merger Regulation) and to prevent concentrations that would significantly impede effective competition in the EEA or any substantial part of it.

The vast majority of notified mergers do not pose competition problems and are cleared after a routine review. From the moment a transaction is notified, the Commission generally has 25 working days to decide whether to grant approval (Phase I) or to start an in-depth investigation (Phase II).

There are currently four on-going Phase II merger investigations: (i) the proposed acquisition of eTraveli by Booking, (ii) the proposed acquisition of Asiana Airlines by Korean Air, (iii) the proposed creation of a joint venture by Orange and MasMovil; and (iv) the proposed acquisition of iRobot by Amazon.

More information will be available on the Commission's competition website, in the Commission's public case register under the case number M.10806.

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Quotes:

> Broadcom holds a very strong position in the market for the supply of certain hardware components. VMware is a key server virtualisation software provider. By acquiring VMware, Broadcom could restrict or degrade interoperability between VMware's leading server virtualization software and some competing hardware components. But the commitments offered by Broadcom will enable its only rival Marvell, to continue competing on equal footing and ensure a similar protection for any future entrants. Margrethe Vestager, Executive Vice-President in charge of competition policy - 12/07/2023

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